THE GYSIN HOMOMORPHISM FACTORS THROUGH RATIONAL EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The Gysin homomorphism attached to a triple, and its behaviour under the proper pushforward of the base. Same chapter chow.tex, labels definition-gysin-homomorphism (statement, proof line 5424), remark-pullback-pairs (proof line 5442) and lemma-closed-in-X-gysin (proof line 5443). Fix the operation being descended and give the compatibility i^* p_* = q_* (i')^* for the base change along the proper morphism used to reduce to a single integral scheme.
- Rational equivalence is preserved by a proper pushforward, and capping with a first Chern class factors through it. Same chapter chow.tex, labels lemma-proper-pushforward-rational-equivalence (proof line 5445) and lemma-factors (proof line 5454). The first lets the reduction to the disjoint union be validated, and the second shows that in the case where the divisor is all of X the value of the Gysin map is rationally trivial.
- The Gysin pullback of a principal divisor on an integral scheme, and additivity of capping with a first Chern class. Same chapter chow.tex, labels lemma-gysin-factors-general (proof line 5457) and lemma-c1-cap-additive (proof line 5459). The first identifies the Gysin pullback of the divisor of a rational function with the cap of the first Chern class of the structure sheaf of the divisor with the fundamental class, and the second observes that this cap is the zero map.

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Source reference. Original definition 02T8: sources/stacks-chow/pinned-original-source.tex, source lines 5069–5069. Navigation pointer for \ref{definition-gysin-homomorphism} used inside the counted span (line 5424); the argument that consumes it is retained in the span.

Source reference. Original lemma 02SP: sources/stacks-chow/pinned-original-source.tex, source lines 4330–4330. Navigation pointer for \ref{lemma-c1-cap-additive} used inside the counted span (line 5459); the argument that consumes it is retained in the span.

Source reference. Original lemma 02TA: sources/stacks-chow/pinned-original-source.tex, source lines 5264–5264. Navigation pointer for \ref{lemma-closed-in-X-gysin} used inside the counted span (line 5443); the argument that consumes it is retained in the span.

Source reference. Original lemma 02TI: sources/stacks-chow/pinned-original-source.tex, source lines 4948–4948. Navigation pointer for \ref{lemma-factors} used inside the counted span (line 5454); the argument that consumes it is retained in the span.

Source reference. Original lemma 02TM: sources/stacks-chow/pinned-original-source.tex, source lines 5352–5352. Navigation pointer for \ref{lemma-gysin-factors-general} used inside the counted span (line 5457); the argument that consumes it is retained in the span.

Source reference. Original lemma 02S2: sources/stacks-chow/pinned-original-source.tex, source lines 3217–3217. Navigation pointer for \ref{lemma-proper-pushforward-rational-equivalence} used inside the counted span (line 5445); the argument that consumes it is retained in the span.

Source reference. Original remark 0B6Y: sources/stacks-chow/pinned-original-source.tex, source lines 5248–5248. Navigation pointer for \ref{remark-pullback-pairs} used inside the counted span (line 5442); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 5421); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $(\mathcal{L}, s, i : D \to X)$ be as in Definition 02T8. The Gysin homomorphism factors through rational equivalence to give a map $i^* : \mathrm{CH}_{k+1}(X) \to \mathrm{CH}_k(D)$.*

Proof. Let $\alpha \in Z_{k+1}(X)$ and assume that $\alpha \sim_{rat} 0$. This means there exists a locally finite collection of integral closed subschemes $W_j \subset X$ of δ-dimension $k+2$ and $f_j \in R(W_j)^*$ such that $\alpha = \sum i_{j,*}\mathrm{div}_{W_j}(f_j)$. Set $X' = \coprod W_i$ and consider the diagram

$$\begin{array}{ccc} D' & \xrightarrow{i'} & X' \\ {\scriptstyle q}\downarrow & & \downarrow{\scriptstyle p} \\ D & \xrightarrow{i} & X \end{array}$$

of Remark 0B6Y. Since $X' \to X$ is proper we see that $i^*p_* = q_*(i')^*$ by Lemma 02TA. As we know that q_* factors through rational equivalence (Lemma 02S2), it suffices to prove the result for $\alpha' = \sum \mathrm{div}_{W_j}(f_j)$ on X'. Clearly this reduces us to the case where X is integral and $\alpha = \mathrm{div}(f)$ for some $f \in R(X)^*$.

Assume X is integral and $\alpha = \mathrm{div}(f)$ for some $f \in R(X)^*$. If $X = D$, then we see that $i^*\alpha$ is equal to $c_1(\mathcal{L}) \cap \alpha$. This is rationally equivalent to zero by Lemma 02TI.

If $D \neq X$, then we see that $i^*\mathrm{div}_X(f)$ is equal to $c_1(\mathcal{O}_D) \cap [D]_{n-1}$ in $\mathrm{CH}_{n-2}(D)$ by Lemma 02TM. Of course capping with $c_1(\mathcal{O}_D)$ is the zero map (Lemma 02SP). $\square$